

Code

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```
# read in packages
library(tidyverse)
library(here)
library(janitor)
library(lubridate)
library(showtext)
library(treemap)
library(viridis)
library(ggribes)

# read in my data and store it as an object called ben-data
ben_data <- read.csv(here("data", "ben-data-tracker - Sheet1.csv"))
```

Data Cleaning

```
# create new object called ben_clean from ben_data
ben_clean <- ben_data |>
  # remove capital letters, add underscores
  clean_names() |>
  # convert dates into an ordered variable
  mutate(date_clean = ymd(date)) |>
  # set day of the week to have levels by chronological order
  mutate(day_of_week = as_factor(day_of_week),
         day_of_week = fct_relevel(day_of_week,
                                   "Monday",
                                   "Tuesday",
                                   "Wednesday",
                                   "Thursday",
                                   "Friday",
                                   "Saturday",
                                   "Sunday")) |>

  # remove unneeded columns
  subset(select = -date)

# check data frame's structure to ensure columns are read correctly
str(ben_clean)
```

```
'data.frame':  32 obs. of  7 variables:
 $ distance_walked_km: num  8.03 6.23 4.49 4.29 7.04 ...
 $ day_of_week       : Factor w/ 7 levels "Monday","Tuesday",...: 4 5 6 7 1 2 3 4 5 6 ...
 $ bus_trips         : int   3 0 0 0 0 2 2 3 2 0 ...
 $ calendar_events   : int   5 4 1 0 3 6 4 4 4 1 ...
 $ high_temperature_c: int   22 21 16 18 19 21 21 22 19 18 ...
 $ lunch_location    : chr   "IV" "Campus" "IV" "Campus" ...
 $ date_clean        : Date, format: "2026-04-23" "2026-04-24" ...
```

```
# add custom fonts
font_add_google("Darker Grotesque", "grot")
font_add_google("Lexend", "lex")
# use showtext to render fonts
showtext_auto()
```

Day of week and date now have associated number values, so the data can be ordered. Each figure will use different columns, so I will begin each plot with a select function.

Figure 1 - Timeseries

Variables being used: distance walked, date, day of week

The aim of this time series is to provide a general summary of how much I walked each day over the course of data collection. Furthermore, I highlighted each day of the week to see if any patterns emerged (for example, if there was difference between on walking on weekdays vs weekends).

```
# create new object to only include needed columns
ben_time <- ben_clean |>
  # select needed columns
  select(date_clean, distance_walked_km, day_of_week)
```

```
# add base layer using cleaned data for this chart
# x-axis: date
# y-axis: distance walked
# color: by day of week (for points)
ggplot(data = ben_time,
       mapping = aes(
         x = date_clean,
         y = distance_walked_km,
         color = day_of_week
       )) +
  # add the distance walked information
  geom_line(
    # use the distance walked column
    mapping = aes(y = distance_walked_km),
    # grey colored line
    color = "grey10",
    # adjust the line's width
    linewidth = 0.3
  ) +
  # add points
  geom_point() +
  # color points in rainbow order by day of week
  scale_color_manual(values = c(
    "Monday" = "red",
    "Tuesday" = "orange2",
    "Wednesday" = "yellow",
    "Thursday" = "green4",
    "Friday" = "cyan3",
```

```

    "Saturday" = "blue1",
    "Sunday" = "purple3"
  )) +
  # adjust y-axis to start at 0
  ylim(0, 14) +
  # adjust color of the background and panel grid to be the same
  # hides grid
  theme(
    panel.background = element_rect(fill = "#D1BCA1"),
    panel.grid = element_line(color = "#D1BCA1")) +
  # creating title
  # renaming axes
  # renaming legend (mapped to color)
  labs(
    x = "Date",
    y = "Distance Walked (km)",
    color = "Day Of Week",
    title = "Daily Distance Walked"
  ) +
  # add an annotation
  annotate(
    # annotation uses curve geometry
    geom = "curve",
    # sets the beginning x location of the curve (must use date format to
    # align with chart)
    x = as.Date("2026-05-08"),
    # sets the beginning y location of the curve
    y = 11.25,
    # sets the end x location of the curve
    xend = as.Date("2026-05-15"),
    # sets the end y location of the curve
    yend = 13.13,
    # sets direction of curve
    curvature = -0.5,
    # adjusts arrow length
    arrow = arrow(length = unit(4, "mm"))) +
  # add text annotation
  annotate(
    # use text geometry
    geom = "text",
    # x location of text
    x = as.Date("2026-05-08"),

```

```

# y location of text
y = 11,
# actual text
label = "Hiking Day",
# center text around set x location
hjust = "center",
# use larger size
size = 8,
# use custom font
family = "grot") +
# annotation 2, same format as before, so fewer annotations
# curve and arrow geometry
annotate(
  geom = "curve",
  x = as.Date("2026-05-17"),
  y = 2.5,
  xend = as.Date("2026-05-11"),
  yend = 3.33,
  curvature = 0.5,
  arrow = arrow(length = unit(4, "mm"))) +
annotate(
  geom = "text",
  x = as.Date("2026-05-17"),
  y = 1,
  # actual text with line break
  label = "Less walking \non Sundays (usually)",
  hjust = "center",
  size = 5,
  family = "grot")

```

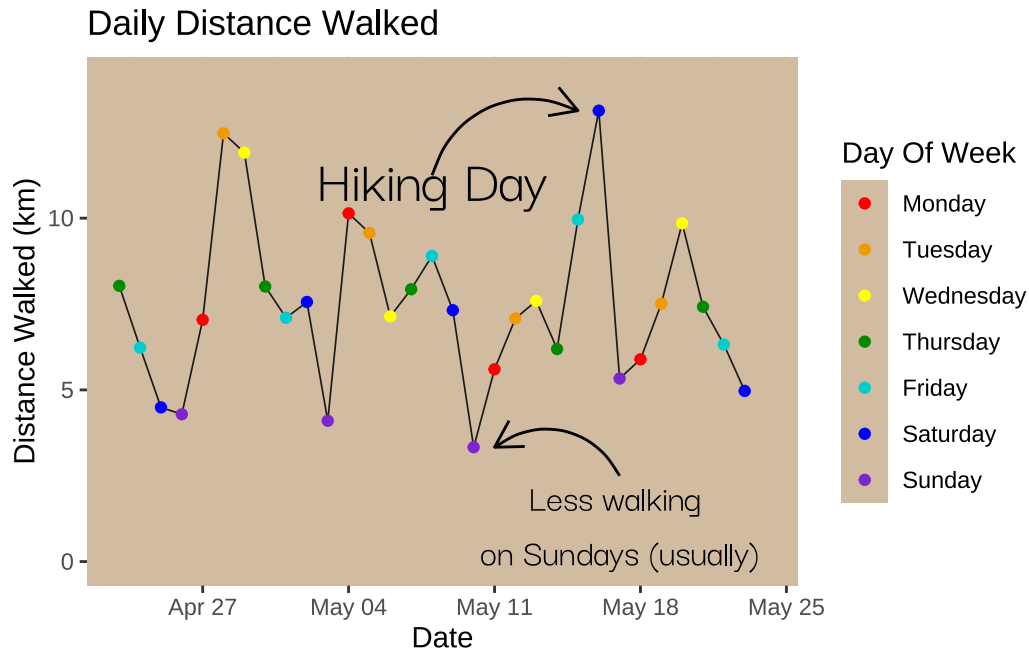


Figure 2 - Treemap

Variables being used: distance walked, bus trips, lunch location

Although distance walked dictates the size of each box, the time series shows that each day does not differ too much in terms of distance walked, and this chart also shows that the sizes are more or less homogeneous when averaged out, visually showing that lunch location and bus trips did not alter distance walked that much. What is interesting is how often I ate lunch on campus. I am busy after all, and that meal plan sure comes in handy.

```
# create new object to only include needed columns
ben_tree <- ben_clean |>
  # select needed columns
  select(distance_walked_km, bus_trips, lunch_location)

# create a treemap diagram
# use ben_tree for the data
treemap(ben_tree,
  # hierarchy of variables to display
  index = c("lunch_location", "bus_trips", "distance_walked_km"),
  # box size controlled by distance walked
  vSize = "distance_walked_km",
```

```

# creates plot title
title = "Distance Walked (km) by Lunch Location (black) and Bus Trips (white)",
fontsize.title = 12,
# sets the color palette based on layer 1 (lunch location)
palette = c("#2F2ABF", "#2C6B36", "#C21D1D", "#BFAB1B"),
# sets border colors for the 3 layers
# black for first 2, grey for 3rd
border.col=c("black", "black", "grey50"),
# adjust border widths, higher layers are thicker
border.lwds=c(4,2.5,1),
# adjusts font size on labels, 3rd layer has no labels since size already
# accounts for the value
fontsize.labels=c(20, 15, 0),
# sets layer 1 text to black and layer 2 to white
fontcolor.labels=c("black", "white"),
# removes highlighting on labels
bg.labels=c("transparent"),
# layer 1's labels are centered, layer 2 in top left of box
align.labels=list( c("center", "center"), c("left", "top")),
fontfamily.labels = "lex",
)

```

Distance Walked (km) by Lunch Location (black) and Bus Trips (white)

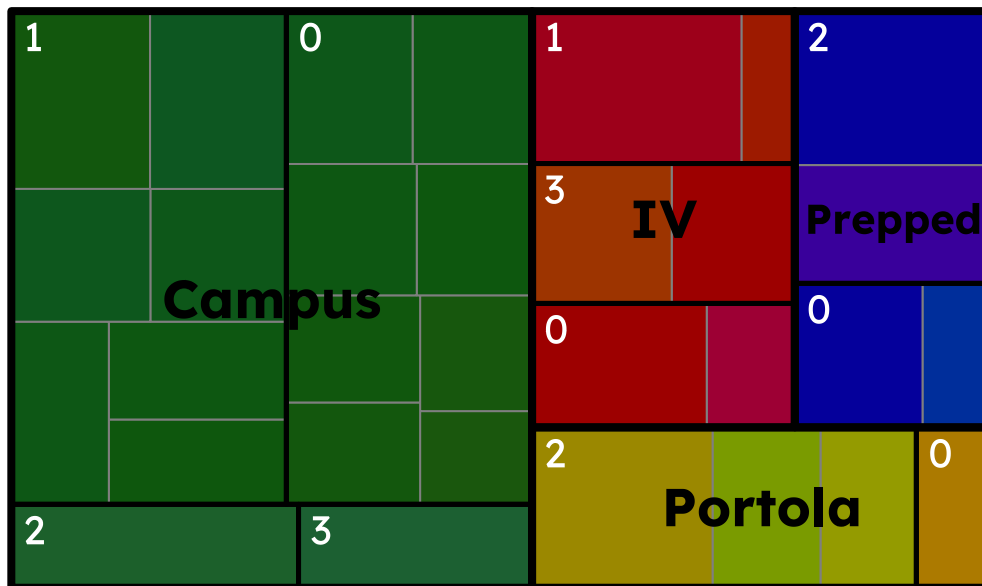


Figure 3 - Scatterplot

Variables used: distance_walked, day_of_week, calendar_events

The scatterplot finally has a clear pattern emerging. There is a small pattern of me walking less on Sundays, which the time series already illustrated, but even more striking is how many fewer calendar events I had on weekends. This makes sense in context, but I am surprised by how clear it is on the chart. Aside from Memorial Day (which is essentially a weekend), weekends have strictly fewer calendar events than weekdays (absolutely no exceptions here). This pattern is notable enough that I drew attention to it.

```
# create new object to only include needed columns
ben_scatter <- ben_clean |>
  # select needed columns
  select(distance_walked_km, day_of_week, calendar_events)
```

```
# create a base layer for a scatterplot
# data: use ben_scatter
# x-axis: calendar events
# y-axis: distance walked
# color: group by day of week
ggplot(
  data = ben_scatter,
  mapping = aes(
    x = calendar_events,
    y = distance_walked_km,
    color = day_of_week
  )) +
  # create scatterplot
  geom_point() +
  # adjust colors, same as figure 1
  scale_color_manual(values = c(
    "Monday" = "red",
    "Tuesday" = "orange2",
    "Wednesday" = "yellow",
    "Thursday" = "green4",
    "Friday" = "cyan3",
    "Saturday" = "blue1",
    "Sunday" = "purple3"
  )) +
  # adjust background color and remove panel gridlines
  theme(
```

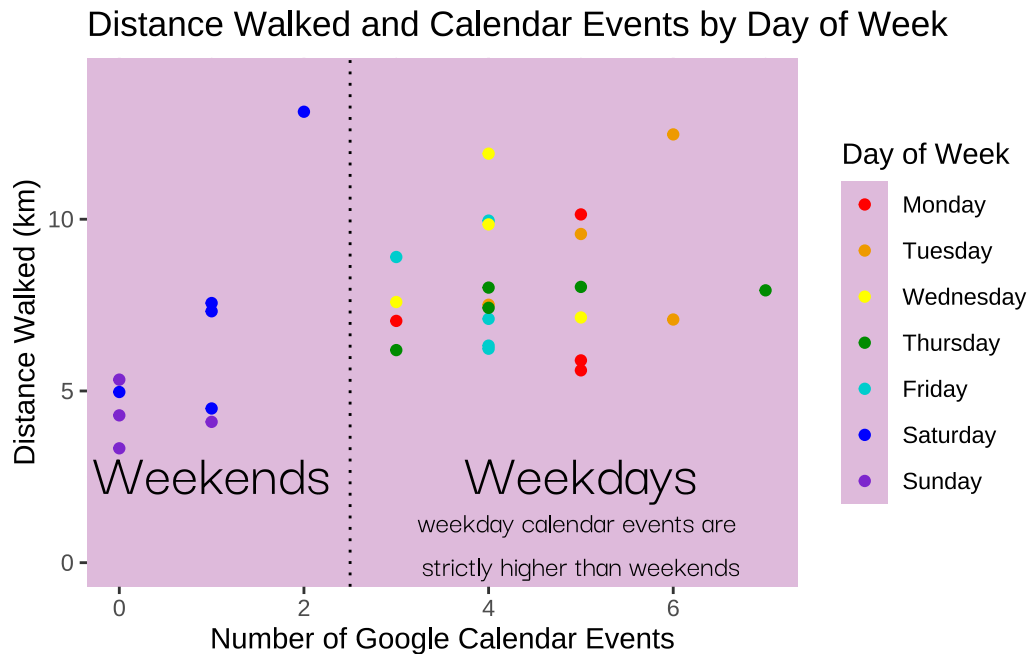


```

    panel.background = element_rect(fill = "#DEBADB"),
    panel.grid = element_line(color = "#DEBADB")) +
# rename axes and legend, create title
labs(
  x = "Number of Google Calendar Events",
  y = "Distance Walked (km)",
  title = "Distance Walked and Calendar Events by Day of Week",
  color = "Day of Week"
) +
# adjust y-axis limits so it starts at 0
ylim(0, 14) +
# annotate with a vertical line to highlight weekday vs weekend pattern
# xintercept sets the location
# color sets the color to black
# linetype makes it dotted
geom_vline(
  xintercept = 2.5,
  color = "black",
  linetype = "dotted"
) +
# since I annotated these in detail in figure 1, I'll make them sparse here
# creating the weekend annotation
annotate(
  geom = "text",
  x = 1,
  y = 2.5,
  label = "Weekends",
  hjust = "center",
  size = 8,
  family = "grot") +
# create annotation for weekday label
annotate(
  geom = "text",
  x = 5,
  y = 2.5,
  label = "Weekdays",
  hjust = "center",
  size = 8,
  family = "grot") +
# create annotation describing the pattern
annotate(
  geom = "text",

```

```
x = 5,
y = 0.5,
# line break needed
label = "weekday calendar events are \nstrictly higher than weekends",
hjust = "center",
size = 4,
family = "grot")
```



Final Write Up

Sources of Error

It turns out that satellite tracking on phones is not perfect, as I noticed the app had me in the wrong location sometimes and would then correct and add a few hundredths of a mile here and there. This means there was a systemic error, but it seemed to amount to less than half of a kilometer per day. I was more interested in patterns rather than absolute walking distance, so this error can mostly be ignored, but is worth noting.

There were also instances where I either forgot to turn the app off (ex: if I was carpooling somewhere) or forgot to turn it back on. In these instances, I used Google Maps to estimate the distance to correct for it. I also had a friend give me an estimate of distance for one day

with their fitness tracker, as the app was consuming too much battery life, and I didn't have access to charging when I was camping for one weekend.

Figure Creation Challenges

One of the biggest challenges that arose came from working with date formatting in figure 1. By default, Rstudio treated these as categorical variables without order when I needed order for the time series. This required wrangling, and it also messed with the annotations. Part of the annotation requires specifying the x-axis location, and when the x-axis is in date format, the annotation location must also awkwardly be created.

[Wait, this is part 4, I don't need this yet, I will finish it later]